

Online Retail Sales Forecasting and Customer Insights System

Submitted by: Lavanya R

Data Science Batch - 01

Prime Vector Pvt Ltd

Project Overview

This project analyzes online retail transaction data to identify sales trends, customer behavior, product performance, and business insights. The project includes data preprocessing, exploratory data analysis, KPI analysis, RFM-based customer segmentation, and machine learning-based sales forecasting. Multiple regression models were evaluated, followed by a seven-day sales forecast using the selected Random Forest model.

Dataset & Data Preprocessing

Include:

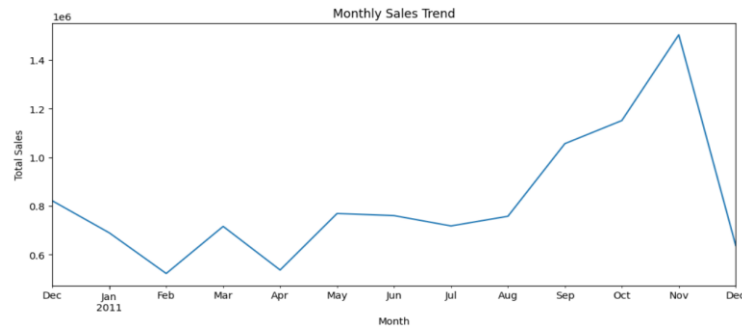
- Original dataset: **541,909 rows × 8 columns**
- Final dataset: **524,878 rows × 11 columns**
- Date range: **Dec 1, 2010 – Dec 9, 2011**

Preprocessing steps:

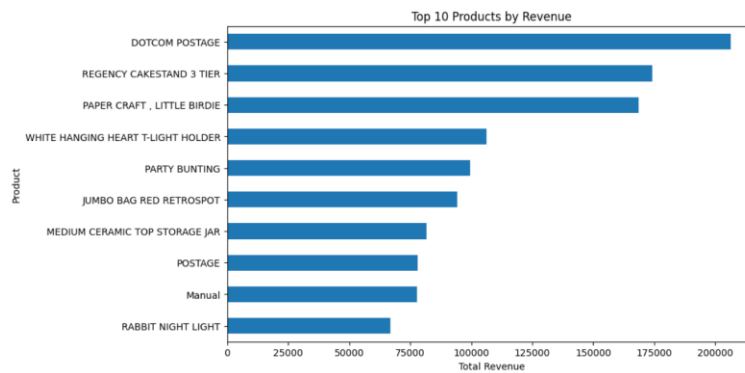
- Removed duplicate records.
- Removed invalid quantities and prices.
- Removed cancelled invoices.
- Created $\text{TotalSales} = \text{Quantity} \times \text{UnitPrice}$.
- Created date-based features.
- Created a separate dataset for customer analysis.

Exploratory Data Analysis

Monthly Sales Trend

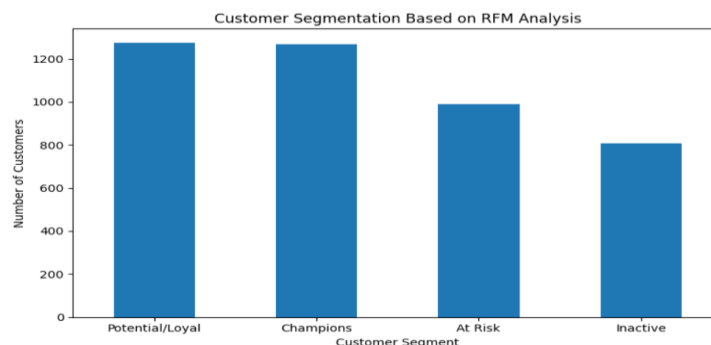


Top Products



Customer Analysis

RFM analysis was used to evaluate customers based on Recency, Frequency, and Monetary value. Customers were assigned quartile-based scores and grouped into business-oriented segments.



Segment	Customers
Champions	1,267
Potential/Loyal	1,276
At Risk	989
Inactive	806

KPI & Statistical Analysis

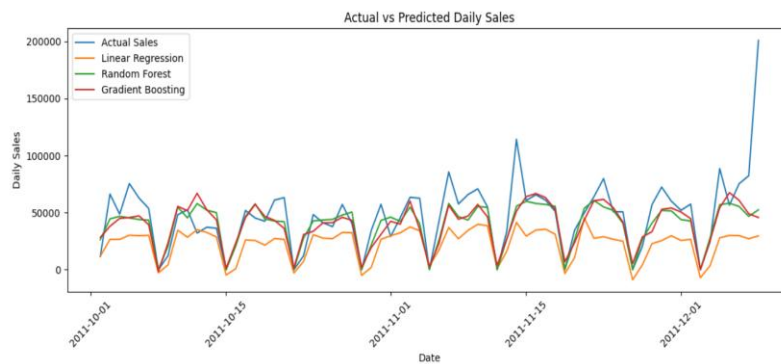
KPI	Value
Revenue	₹10.64M
Orders	19,960
Customers	4,338
Products	3,922
Avg Order Value	₹533.17

The mean order revenue of ₹533.17 was higher than the median of ₹303.30, indicating a right-skewed distribution with some high-value orders.

Sales Forecasting & Model Evaluation

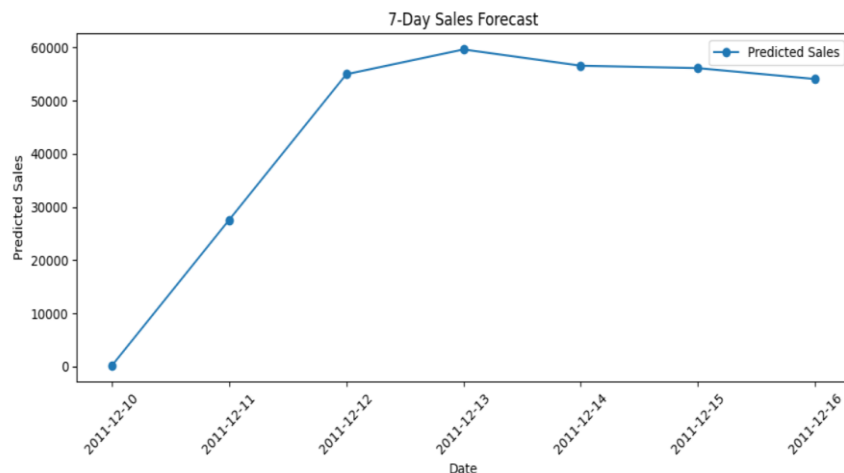
Daily sales were aggregated and calendar, lag, and rolling-average features were created. An 80:20 chronological split was used for training and testing.

Model	MAE	RMSE	R ²
Linear Regression	25,273.72	34,520.07	-0.2196
Gradient Boosting	14,139.41	24,857.35	0.3676
Random Forest	13,117.32	23,491.33	0.4352



Future Forecast

The model generated a seven-day forecast with total predicted sales of **₹309,214.22** and average predicted daily sales of **₹44,173.46**. The near-zero prediction for December 10 reflects the historical dataset's consistent zero-sales pattern on Saturdays.



Business Insights & Recommendations

Insights

- November 2011 had the highest complete monthly revenue.
- UK generated the majority of recorded revenue.
- 41.4% of known customers were classified as At Risk or Inactive.
- Weekly patterns were important in sales forecasting.

Recommendations

- Develop retention campaigns for At Risk/Inactive customers.
- Use loyalty strategies for high-value customers.
- Use forecasts for inventory and demand planning.
- Improve customer identification/data quality.
- Retrain the forecasting model as new sales data becomes available.

Conclusion

The project successfully combined retail data analysis, customer segmentation, statistical analysis, and machine learning forecasting to generate actionable business insights. The Random Forest model achieved an R^2 of 0.4352 on the held-out test period and was used to generate a seven-day forecast. The analysis demonstrates how historical transaction data can support customer retention, sales planning, and data-driven decision-making.